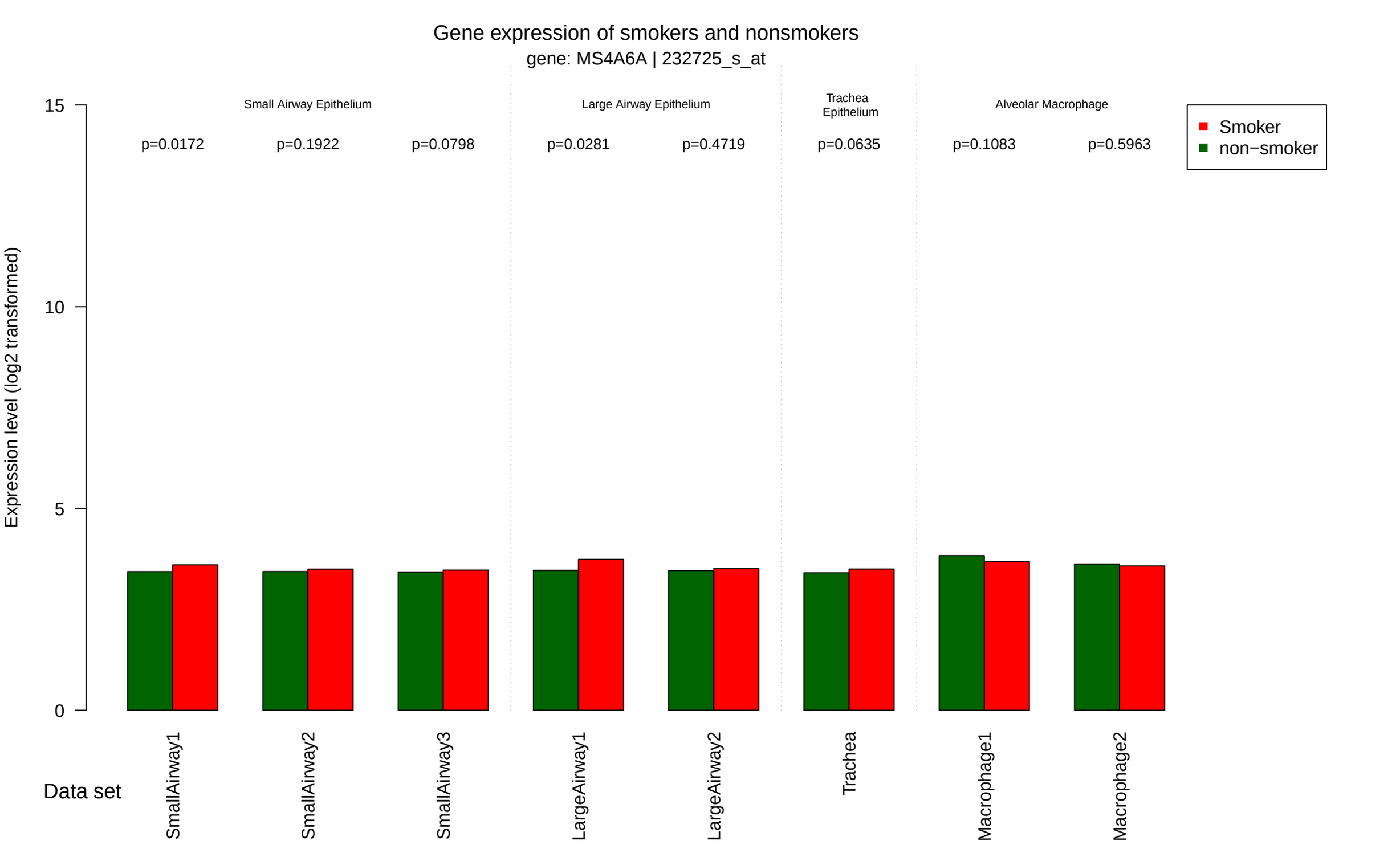
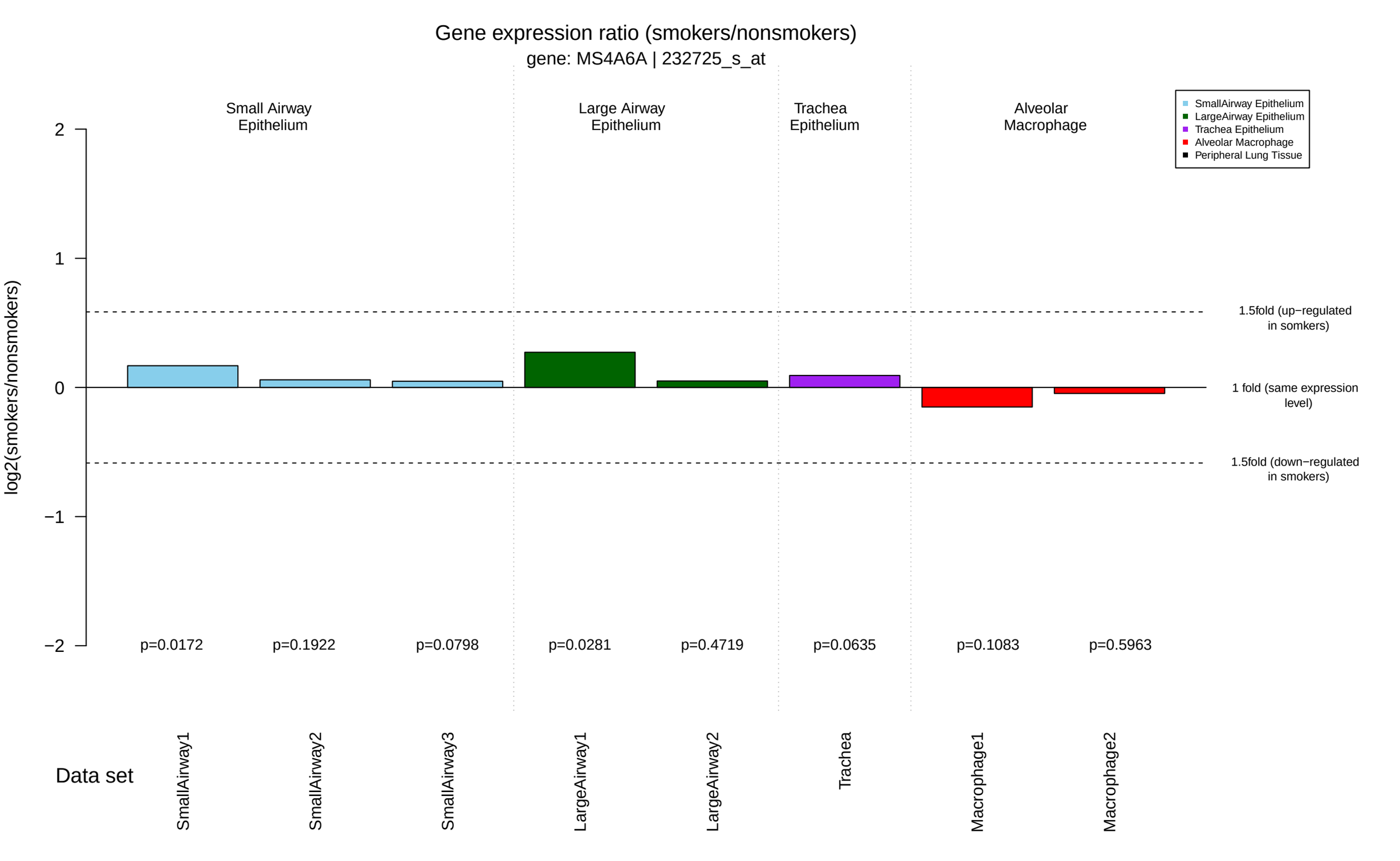
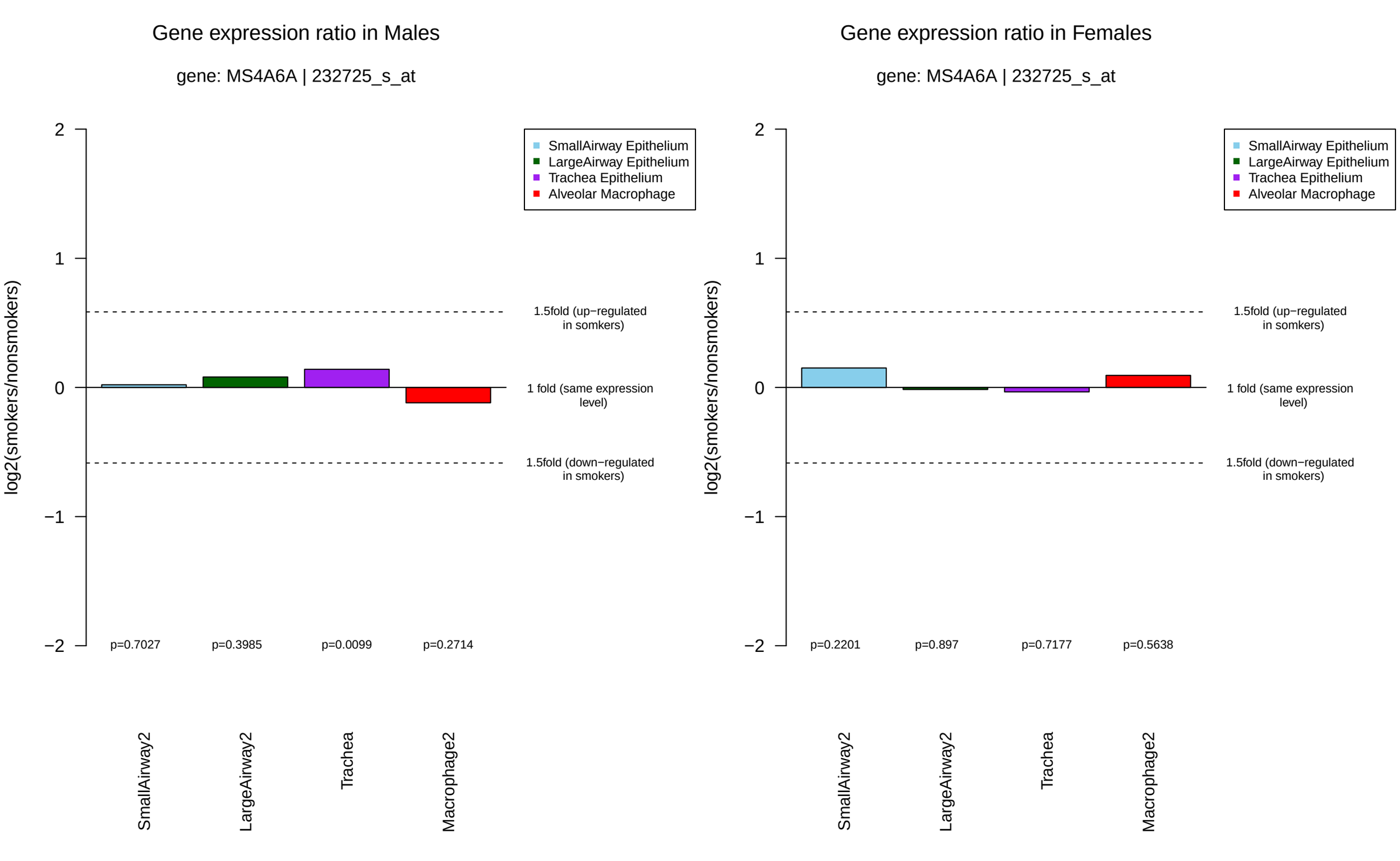
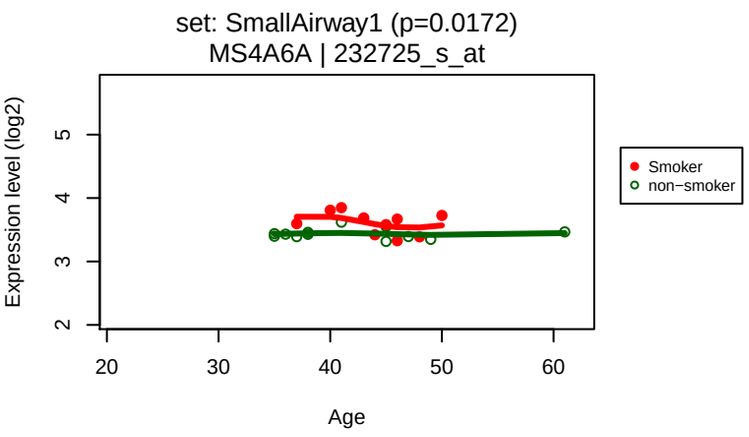
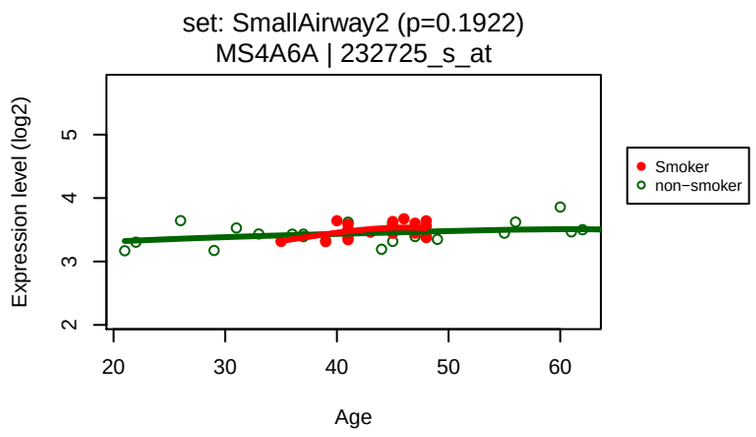
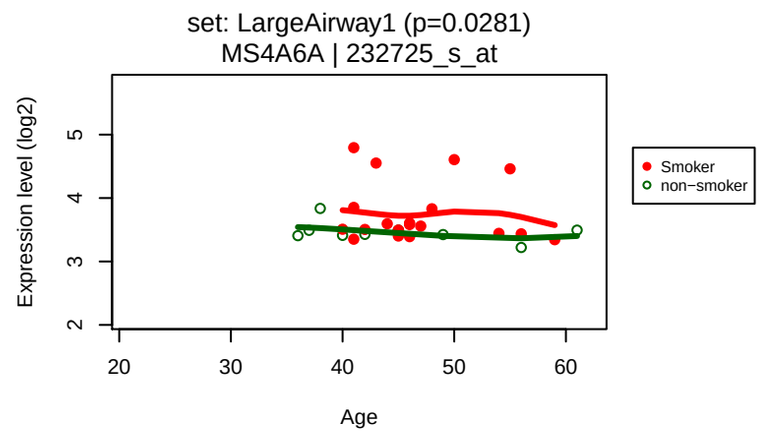

Gene expression of smokers and nonsmokers
gene: MS4A6A | 232725_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0172
p=0.1922
p=0.0798
p=0.0281
p=0.4719
p=0.0635
p=0.1083
p=0.5963
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

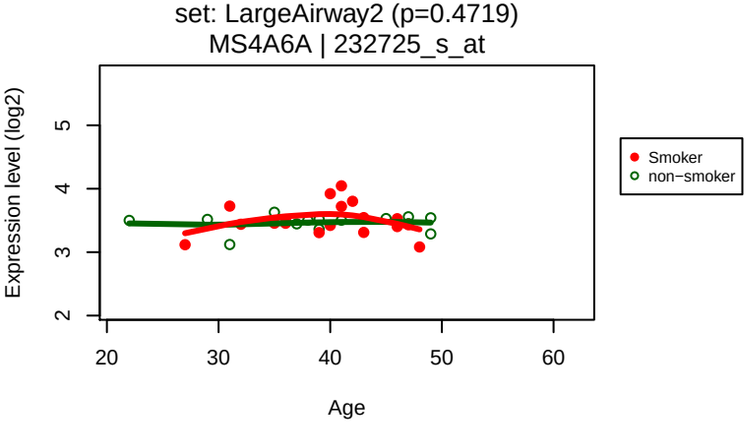

Gene expression ratio (smokers/nonsmokers)
gene: MS4A6A | 232725_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0172
p=0.1922
p=0.0798
p=0.0281
p=0.4719
p=0.0635
p=0.1083
p=0.5963
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

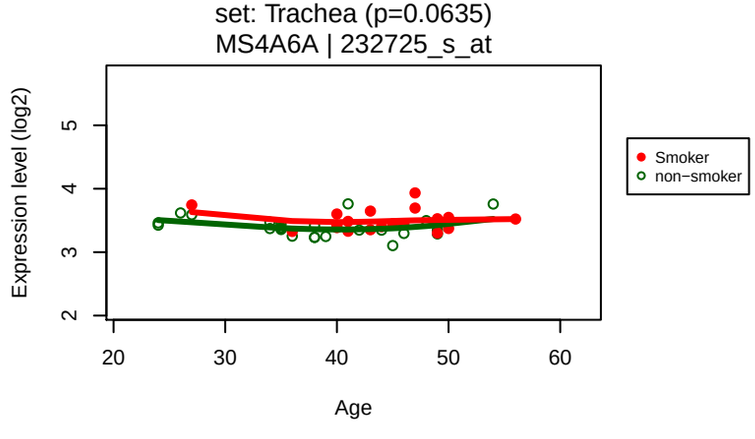

Gene expression ratio in Males
gene: MS4A6A | 232725_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7027
p=0.3985
p=0.0099
p=0.2714
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MS4A6A | 232725_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2201
p=0.897
p=0.7177
p=0.5638
SmallAirway2
LargeAirway2
Trachea
Macrophage2

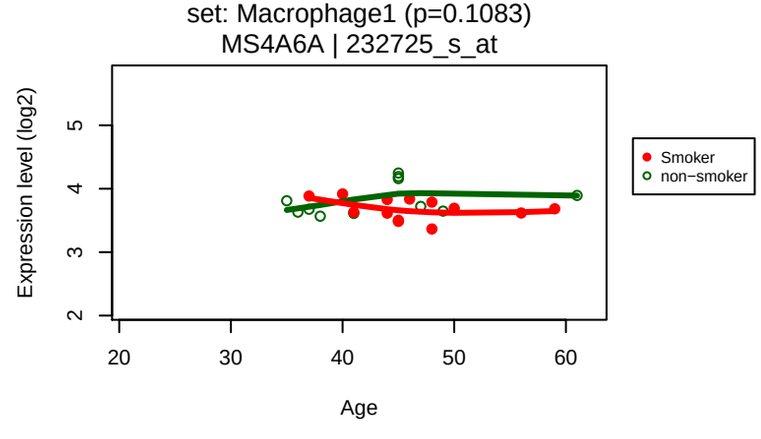

set: SmallAirway1 (p=0.0172)
MS4A6A | 232725_s_at
Expression level (log2)
Age
Smoker
non−smoker

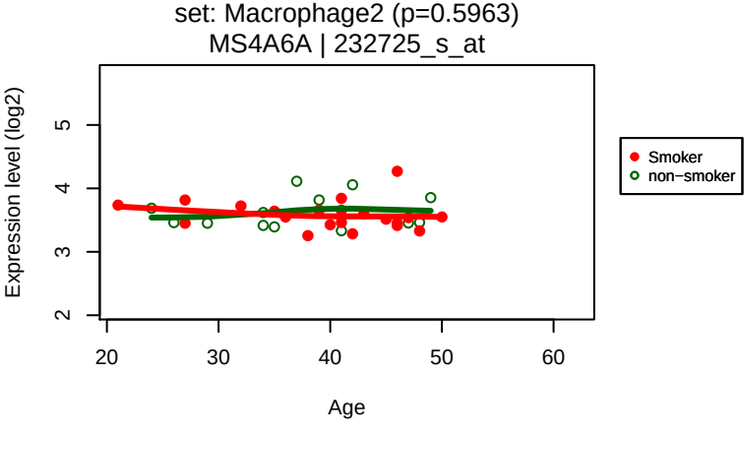

set: SmallAirway2 (p=0.1922)
MS4A6A | 232725_s_at
Expression level (log2)
Age
Smoker
non−smoker

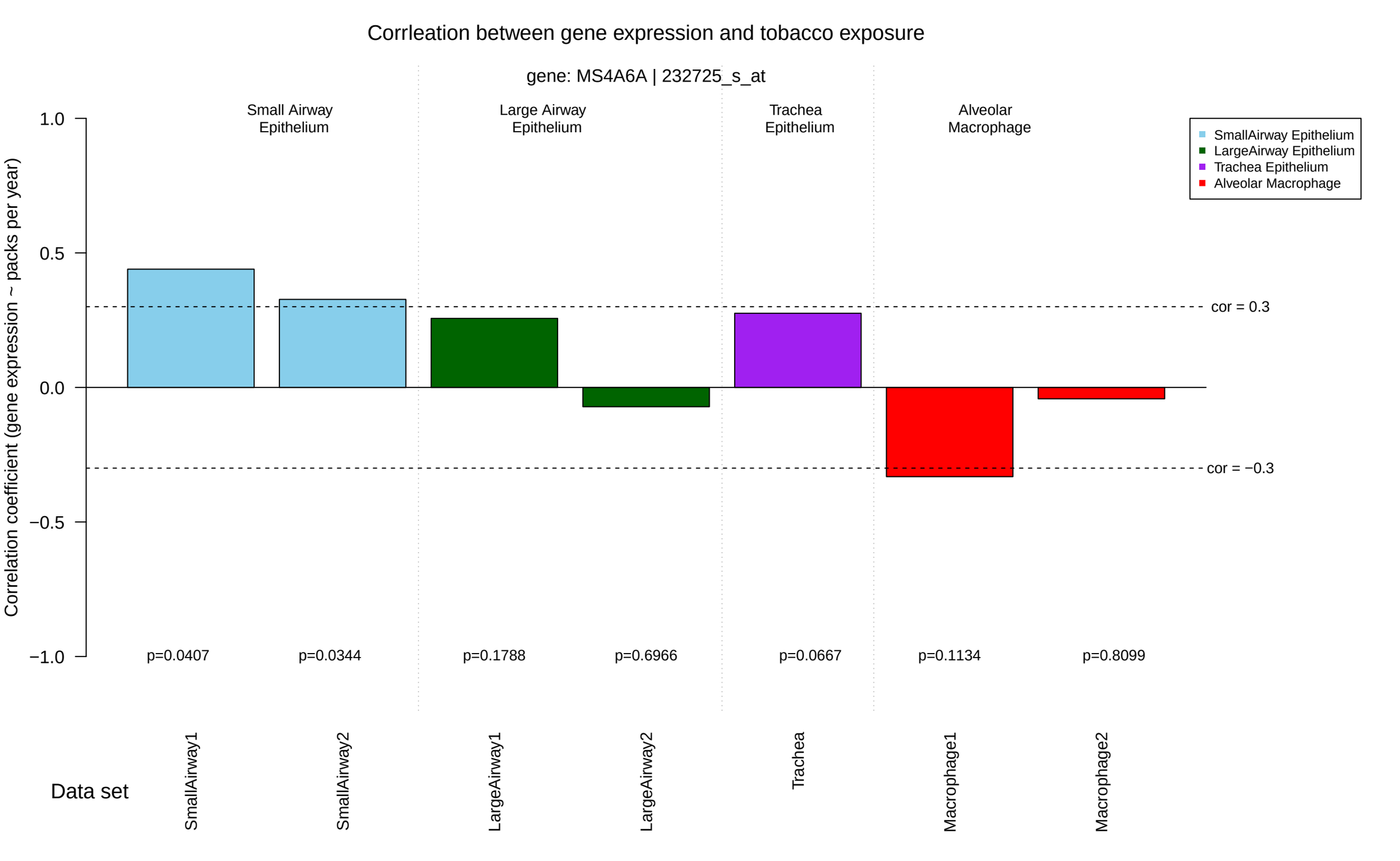

set: LargeAirway1 (p=0.0281)
MS4A6A | 232725_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4719)
MS4A6A | 232725_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0635)
MS4A6A | 232725_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1083)
MS4A6A | 232725_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5963)
MS4A6A | 232725_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MS4A6A | 232725_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0407
p=0.0344
p=0.1788
p=0.6966
p=0.0667
p=0.1134
p=0.8099
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2